THE TECHNICAL UNIVERSITY OF KENYA

School of Computing and Information Technology | Department of Computer Science & Informatics

PRESENTATION SLIDES DEFENSE COMPANION
SPEAKER SCRIPTS & TIMING MANUAL

Dedicated Slide-by-Slide Walkthrough for Project A (15 Mins) & Project B (20 Mins)

Candidate & Reg No:	Owuor Collins | SCCI/01227/2022 (B.Tech Information Technology)
Supervisors:	Dr. Edwin Ngwawe & Mr. Peter Maina Kariuki
Time Allocations:	Project A Proposal: Exactly 15 Minutes (16 Slides) | Project B Final: Exactly 20 Minutes (20 Slides)

PRESENTATION PACING & TIMING MASTERY

Examiners penalize presenters who exceed their allocated time or rush frantically through slides. Keep this exact pacing mindset:

- **Project A Defense (15 Minutes):** 16 slides total. Allocate ~50 to 55 seconds per slide. Focus heavily on Problem Statement, Objectives, Scope, and Feasibility.
- **Project B Final Defense (20 Minutes):** 20 slides total. Allocate ~60 seconds per slide. Focus heavily on Architecture, Core Modules Implemented, Testing, and UAT Impact.
- **Golden Delivery Rule:** Never read bullet points off the slide. Bullets are visual summaries for the panel. Say the spoken scripts provided in this guide.

PART 1: PROJECT A PRESENTATION — 15-MINUTE DEFENSE (16 SLIDES)

Target Duration: 15 Minutes (16 Slides | ~50 to 55 Seconds per Slide). Focus: Problem, Feasibility, Plan, Scope, and Requirements.

Slide 1: Title Slide (Project Proposal Presentation) [00:00 – 00:45 (45s)]

- • **Slide Content:** Project Title: Laboratory Information Management System for Yala Sub-County Hospital; Candidate: Owuor Collins; Supervisors: Dr. Edwin Ngwawe & Mr. Peter Maina Kariuki.

> **What to Say to the Panel (Speaker Script):**
> *"Good morning respected members of the panel, my supervisors Dr. Edwin Ngwawe and Mr. Peter Maina Kariuki. My name is Owuor Collins, registration number SCCI/01227/2022. Today I am presenting my Project A proposal: Development of a Web-Based Laboratory Information Management System for Yala Sub-County Hospital."*

> **Defense Strategy & Examiner Tip:**
> *Speak with steady voice, formal greeting, and acknowledge your supervisors.*

Slide 2: Introduction & National Context [00:45 – 01:40 (55s)]

- • **Slide Content:** Digital health transition under Kenya eHealth Policy; Yala Hospital profile (Level 4 in Gem Sub-County); 70% of medical diagnoses depend directly on laboratory tests; manual registers cause delays and lost slips.

> **What to Say to the Panel (Speaker Script):**
> *"Healthcare delivery in Kenya is rapidly aligning with the National Digital Health Strategy. Crucially, over 70% of clinical medical decisions depend directly on accurate laboratory diagnostic tests. At Yala Sub-County Hospital, a busy Level 4 facility, the laboratory remains entirely paper-dependent. Our project proposes a localized, web-based LIMS to eliminate bottlenecks and bridge this digital divide."*

> **Defense Strategy & Examiner Tip:**
> *Highlight the '70% clinical dependency' fact—it proves why fixing the lab has the highest clinical payoff.*

Slide 3: Background Study: The Operational Environment [01:40 – 02:35 (55s)]

- • **Slide Content:** Level 4 facility on Kisumu-Busia highway; high outpatient, ANC, and emergency trauma volume; physical paper MOH 204 request slips and MOH 240 registers; manual paper transit across open hospital corridors.

What to Say to the Panel (Speaker Script):
"Yala Sub-County Hospital serves a high patient volume along the Kisumu-Busia transit corridor, including routine outpatient, antenatal clinics, and highway trauma emergencies. Currently, doctors write test orders in paper MOH 204 books, and lab technicians record findings in manual MOH 240 registers. Patients physically carry slips between consultation rooms and the lab, leading to damaged slips, lost records, and uncoordinated care."

Defense Strategy & Examiner Tip:
Emphasize physical paper transit between buildings—it explains why slips get lost.

Slide 4: Problem Statement: 6 Measurable Bottlenecks [02:35 – 03:35 (60s)]

- • **Slide Content:** Prolonged turnaround times (3-5 hours); illegible handwriting causing transcription errors; ~12% monthly lost result slips; absence of urgent panic alerts; 2 full days of manual MOH 706 tallying; cash handling violating cashless laws.

What to Say to the Panel (Speaker Script):
"Our field investigation revealed six critical operational failures: First, prolonged turnaround times averaging 3 to 5 hours for basic tests. Second, illegible clinician handwriting causing dangerous transcription errors. Third, approximately 12% of result slips are lost every month, forcing painful repeat blood draws. Fourth, no urgent panic alerts for critical values like severe malaria or critical anemia. Fifth, compiling monthly MOH 706 reports consumes two full working days of manual tallying. Finally, counter cash handling violates public finance cashless laws."

Defense Strategy & Examiner Tip:
State the 12% lost slips and 2-day MOH tallying clearly; these are your core problem metrics.

Slide 5: Project Objectives (General & 4 Specific) [03:35 – 04:30 (55s)]

- • **Slide Content:** General Objective: Design, develop, and evaluate a web-based LIMS. Specific Objectives: 1. Analyze manual bottlenecks; 2. Design 3-tier architecture & RBAC schema; 3. Develop CPOE ordering, test bench, panic alerts & eCitizen billing; 4. Evaluate via test cases and UAT.

What to Say to the Panel (Speaker Script):
"The general objective is to design, develop, and evaluate a web-based LIMS for Yala Sub-County Hospital. We decomposed this into four specific objectives following the software engineering lifecycle: First, analyze manual workflow bottlenecks; second, design a 3-tier architecture, normalized database schema, and role-based access control; third, develop core modules including CPOE ordering, testing bench, and cashless eCitizen

settlement; and fourth, evaluate system performance through 14 test cases and User Acceptance Testing."

Defense Strategy & Examiner Tip:
Point out the software lifecycle structure: Analyze -> Design -> Develop -> Evaluate.

Slide 6: Project Scope: Clear System Boundaries [04:30 – 05:25 (55s)]

- • **Slide Content:** In-Scope: Native PHP 8.2 & MySQL on Apache; 5 RBAC roles; OPD sequencing & Code128 barcode slips; CPOE ordering & testing bench; eCitizen Paybill 222222 billing. Out-of-Scope: Direct RS-232 serial machine cables; live DHIS2 cloud push; physical LAN hospital rewiring.

What to Say to the Panel (Speaker Script):
"To guarantee successful delivery, we clearly defined project scope. In-scope items include our modular PHP 8.2 and MySQL stack, role-based access control across five hospital roles, barcode-enabled routing slips, doctor test ordering, panic alerts, and eCitizen Paybill 222222 integration. We deliberately scoped out direct machine analyzer cabling and national DHIS2 cloud push for Phase 2, ensuring our core system is self-reliant and immune to external hardware and internet failures."

Defense Strategy & Examiner Tip:
Preempt the question 'Did you connect to the physical analyzer machine?' by stating it was bounded to Phase 2.

Slide 7: Justification & Significance [05:25 – 06:15 (50s)]

- • **Slide Content:** Administration: Zero stationery expenses & sealed billing leakage; Clinicians: Instant diagnostic history & clinical allergy banners; Lab Technologists: Clear worklists & automated reference ranges; Patients: Zero lost slips & faster care; Public Health: Accurate disease surveillance.

What to Say to the Panel (Speaker Script):
"This project justifies itself across five hospital stakeholder groups: Hospital administration eliminates recurring stationery costs and stops revenue leakage. Clinicians gain instant access to patient diagnostic records and allergy warnings. Laboratory technologists benefit from automated reference ranges and 1-click reports. Patients experience zero lost slips and significantly reduced wait times. Finally, county health authorities gain accurate epidemiological data."

Defense Strategy & Examiner Tip:

Show that the project benefits clinicians, patients, and administrators alike.

Slide 8: Resource Requirements & Budget [06:15 – 07:10 (55s)]

- • **Slide Content:** Hardware: Core i5 workstation PC, thermal barcode printer, 650VA UPS; Software: PHP 8.2, MySQL 8.0, Apache/XAMPP (Zero license cost); Capital Cost: KES 120,000; Training & Reserve: KES 50,000; Total: ~KES 170,000 with rapid ROI.

What to Say to the Panel (Speaker Script):
"The resource budget is highly cost-effective because the software stack—PHP, MySQL, and Apache—incurs zero annual software licensing costs compared to proprietary systems costing millions. Hardware requirements were limited to approximately KES 120,000 for server hardware and backup UPS units, plus KES 50,000 for staff training and user manuals. This modest investment is fully recovered within months through recovered unbilled tests and eliminated paper ledgers."

Defense Strategy & Examiner Tip:
Emphasize zero licensing costs—open-source software saves public hospitals millions.

Slide 9: Project Schedule & Agile Sprints [07:10 – 08:00 (50s)]

- • **Slide Content:** Agile Development Model across 4 two-week iterative sprint cycles; Sprint 1: Inception & SRS; Sprint 2: Architecture, schema & RBAC; Sprint 3: CPOE, lab bench & eCitizen; Sprint 4: Testing, UAT & deployment.

What to Say to the Panel (Speaker Script):
"The project is structured around an Agile development methodology across four iterative two-week sprint cycles. Sprint 1 covers requirements gathering and workflow observation. Sprint 2 establishes our 3-tier architecture, relational schema, and RBAC login. Sprint 3 develops core modules including CPOE ordering, bench testing, and eCitizen billing. Sprint 4 conducts functional testing, hospital UAT evaluation, and user manual documentation."

Defense Strategy & Examiner Tip:
Explain that Agile was selected because hospital staff feedback continuously refines user interfaces.

Slide 10: Gantt Chart & Academic Project Timeline [08:00 – 08:50 (50s)]

- • **Slide Content:** 186-day total project roadmap spanning November 2025 to August 2026; Milestones: Concept Paper (Dec), Proposal (Jan), Literature Review & Methodology (Feb), System Coding (Apr-Jul), Testing & Deployment (Jul-Aug).

What to Say to the Panel (Speaker Script):
"This slide displays our 186-day project Gantt chart roadmap across the academic year. Initial inception and concept drafting began in November, proposal and methodology chapters were completed by February, system coding spanned April through July, and testing, UAT validation, and deployment were completed in August. This structured schedule ensured steady progression and timely milestone delivery."

Defense Strategy & Examiner Tip:
Walk through the timeline confidently to demonstrate strong project management discipline.

Slide 11: Literature Review: Existing Systems Analysis [08:50 – 09:45 (55s)]

- • **Slide Content:** Manual MOH Paper: Cheap but high error rate & lost slips; OpenMRS & KenyaEMR: Robust for HIV/TB clinics, but heavy Java stack & lacks full lab bench workflows; Commercial LIMS (LabWare): Feature-complete but millions in annual licenses; Cloud SaaS: Fails during rural internet downtime.

What to Say to the Panel (Speaker Script):
"In our literature review, we evaluated existing solutions against public hospital realities. Manual paper is cheap but fragile and inefficient. OpenMRS and KenyaEMR are successful national tools for HIV/TB care, but they are resource-heavy and lack full laboratory accessioning and bench workflows. Commercial solutions like LabWare cost millions of shillings in annual vendor licenses. Cloud-only SaaS solutions fail when rural internet goes down."

Defense Strategy & Examiner Tip:
Demonstrates that you know existing systems like OpenMRS and KenyaEMR and can explain their specific limitations.

Slide 12: Literature Review: Identified Gaps & Proposed Solution [09:45 – 10:40 (55s)]

- • **Slide Content:** Gap 1: Fragmented communication silos; Gap 2: Lack of pre-analytical specimen QA; Gap 3: Rural internet vulnerability; Gap 4: Cash handling non-compliance; Proposed Solution: Offline-capable LAN intranet LIMS with eCitizen 222222 integration and ISO 15189 QA.

What to Say to the Panel (Speaker Script):

> *"We identified four major operational gaps: inter-departmental communication silos, lack of pre-analytical specimen quality assurance, vulnerability to rural internet disruptions, and non-compliance with public cashless directives. Our proposed solution directly solves all four: it connects hospital departments over a local intranet, implements ISO 15189 specimen rejection controls, functions 100% without external internet, and reconciles Government eCitizen Paybill 222222 payments."*

> **Defense Strategy & Examiner Tip:**
> *Mention ISO 15189—it is the international medical laboratory quality standard.*

Slide 13: Methodology & Development Tools Stack [10:40 – 11:35 (55s)]

- • **Slide Content:** Development Model: Agile SDLC; Backend: PHP 8.2 procedural architecture; Database: MySQL 8.0 with InnoDB foreign keys; Frontend: Bootstrap 5 & Chart.js; PDF Reports: FPDF library; IDE & VCS: Visual Studio Code & Git/GitHub.

> **What to Say to the Panel (Speaker Script):**
> *"Our technical stack was chosen for performance, low memory overhead, and long-term maintainability. We selected PHP 8.2 for backend business logic, MySQL 8.0 with InnoDB for relational persistence, Bootstrap 5 for a responsive medical interface, and FPDF for compiling digitally certified PDF reports. Development was tracked with Git version control within Visual Studio Code."*

> **Defense Strategy & Examiner Tip:**
> *Why procedural PHP instead of Laravel? It has lower overhead and hospital IT staff can easily maintain it.*

Slide 14: System Analysis & Feasibility Study [11:35 – 12:35 (60s)]

- • **Slide Content:** Technical Feasibility: Stack runs smoothly on existing hospital core-i3/i5 PCs; Operational Feasibility: Digital screens mirror physical MOH 204/240 register layouts; Economic Feasibility: High ROI via eliminated paper and captured billing; 4 User Profiles: Receptionist, Doctor, Lab Tech, Administrator.

> **What to Say to the Panel (Speaker Script):**
> *"Our feasibility evaluation validated technical, operational, and economic viability. Technically, the system runs smoothly on standard workstation PCs over the local network. Operationally, staff adoption is rapid because our screens mirror familiar physical MOH registers. Economically, the cost savings and recovered billing justify deployment. We defined four distinct user profiles enforcing strict Role-Based Access Control."*

Defense Strategy & Examiner Tip:
Mirroring physical MOH registers solved staff change resistance.

Slide 15: System Requirements Modeling (Functional & Non-Functional) [12:35 – 13:35 (60s)]

- • **Slide Content:** Functional: Barcode patient registration, CPOE test ordering, automated panic alerts, certified PDF reports, eCitizen Paybill 222222 billing; Non-Functional: Bcrypt password hashing, MySQLi prepared statements, CSRF tokens, sub-1.5s LAN response time, automated daily SQL backups.

What to Say to the Panel (Speaker Script):
"Requirements modeling established functional and non-functional specifications. Functionally, the system manages barcode patient registration, CPOE test ordering, automated panic alerting, certified PDF generation, and eCitizen billing. Non-functionally, we enforced enterprise-grade security via Bcrypt hashing, parameterized prepared statements, and CSRF token validation, maintaining sub-1.5 second page load times across the hospital LAN."

Defense Strategy & Examiner Tip:
Highlight MySQLi prepared statements and CSRF protection—examiners test security thoroughly.

Slide 16: Academic References & Proposal Conclusion [13:35 – 14:30 (55s)]

- • **Slide Content:** Kenya National eHealth Policy (2016-2030); Kenya Data Protection Act 2019; Ministry of Health Kenya SOPs for Laboratory Services; ISO 15189:2022 Medical Laboratories Standard; Sommerville & Pressman Software Engineering textbooks.

What to Say to the Panel (Speaker Script):
"Our proposal is grounded in recognized academic literature, Ministry of Health laboratory operating procedures, Kenya's Data Protection Act 2019, and ISO 15189 quality standards. Thank you for your time and attention. I now welcome questions, comments, and constructive critique from the examining panel."

Defense Strategy & Examiner Tip:
End with confidence and poise. Stop speaking and listen carefully to the first question.

PART 2: PROJECT B FINAL PRESENTATION — 20-MINUTE DEFENSE (20 SLIDES)

Target Duration: 20 Minutes (20 Slides | ~60 Seconds per Slide). Focus: Implemented Architecture, Database Schema, 7 Core Modules, 14 Test Cases, UAT Results, and Operational Impact.

Slide 1: Title Slide (Final Project Defense Presentation) [00:00 – 00:50 (50s)]

- • **Slide Content:** Project Title: Laboratory Information Management System for Yala Sub-County Hospital; Final Project Defense Presentation (Project B); Candidate: Owuor Collins (SCCI/01227/2022); Supervisors: Dr. Edwin Ngwawe & Mr. Peter Maina Kariuki.

What to Say to the Panel (Speaker Script):
"Good morning respected members of the panel, my supervisors Dr. Edwin Ngwawe and Mr. Peter Maina Kariuki, and members of faculty. My name is Owuor Collins, registration number SCCI/01227/2022. Today, I am proud to present the final defense of my undergraduate project: Development of a Web-Based Laboratory Information Management System for Yala Sub-County Hospital."

Defense Strategy & Examiner Tip:
State that this is the final project defense and emphasize the practical delivery of a working system.

Slide 2: Introduction: Modernizing Clinical Diagnostics [00:50 – 01:50 (60s)]

- • **Slide Content:** Digital health transition under Kenya eHealth Policy; Yala Hospital profile (Level 4 in Gem Sub-County); 70% of medical diagnoses depend directly on reliable lab tests; pre-system manual registers caused delays and lost slips; delivered solution is a localized intranet LIMS.

What to Say to the Panel (Speaker Script):
"To introduce our research, modernizing public healthcare delivery is a top national priority under Kenya's eHealth Policy. Over 70% of clinical medical diagnoses depend directly on reliable laboratory tests. At Yala Sub-County Hospital, a busy Level 4 facility in Siaya County, the laboratory was heavily burdened by manual paper registers. By engineering and deploying a localized web-based LIMS, we automated the complete diagnostic lifecycle from patient triage to certified PDF test results."

Defense Strategy & Examiner Tip:
Reinforce the 70% clinical dependency statistic—it grounds the entire engineering effort.

Slide 3: Background Study: The Operational Environment [01:50 – 02:50 (60s)]

- • **Slide Content:** Level 4 facility on Kisumu-Busia highway; high outpatient volume, antenatal clinics, emergency trauma; manual MOH 204 slips & MOH 240 ledger books; physical paper transit across open corridors; unreceipted fee leakages.

What to Say to the Panel (Speaker Script):
"Yala Sub-County Hospital handles high patient throughput along the Kisumu-Busia transit corridor, including routine outpatients, antenatal clinics, and emergency road trauma cases. Previously, clinicians wrote test orders in paper MOH 204 books, and lab technicians recorded findings in heavy MOH 240 registers. Slips were manually carried across hospital corridors, leading to misplaced records, zero doctor visibility into pending tests, and unreceipted cash fee leakages."

Defense Strategy & Examiner Tip:
Frame the background as the operational reality you investigated first-hand during hospital visits.

Slide 4: Problem Statement: 6 Measurable Clinical Bottlenecks [02:50 – 03:55 (65s)]

- • **Slide Content:** Prolonged turnaround times (3-5 hours); illegible clinician handwriting causing transcription mistakes; 12% lost result slips forcing repeat pricks; absence of urgent panic alerts for critical results; 2 full days of manual MOH 706 tallying; cash handling violating public finance laws.

What to Say to the Panel (Speaker Script):
"Our baseline investigation identified six critical operational bottlenecks: First, excessive turnaround times due to manual paper routing. Second, transcription mistakes caused by illegible clinician handwriting. Third, an estimated 12% of result slips were lost each month, subjecting patients to redundant, painful repeat blood draws. Fourth, absence of panic alerts when life-threatening values were detected. Fifth, compiling monthly MOH 706 reports consumed two full working days of manual ledger counting. Finally, manual cash handling at hospital counters created revenue leakage and violated public cashless regulations."

Defense Strategy & Examiner Tip:
Make eye contact when stating the 12% lost slips and 2-day MOH tallying—these numbers justify the project.

Slide 5: Project Objectives: General & 4 Specific (100% Achieved) [03:55 – 04:55 (60s)]

- • **Slide Content:** General Objective: Design, develop, and evaluate a web-based LIMS. Specific Objectives: 1. Analyze manual bottlenecks; 2. Design 3-tier architecture & RBAC schema; 3. Develop CPOE ordering, test bench, panic alerts & eCitizen billing; 4. Evaluate via 14 test cases and UAT (4.6/5.0).

What to Say to the Panel (Speaker Script):
"Our general objective was to design, develop, and evaluate a web-based LIMS for Yala Sub-County Hospital. I am pleased to report that 100% of our specific objectives were achieved: First, analyzing existing manual bottlenecks; second, designing a 3-tier architecture and relational schema; third, developing core modules including CPOE ordering, bench testing, panic alerts, and eCitizen billing; and fourth, rigorously evaluating system performance through 14 functional test cases and hospital User Acceptance Testing."

Defense Strategy & Examiner Tip:
Say '100% of our specific objectives were achieved'—it shows confidence and completion.

Slide 6: Project Scope: Delivered System vs. Deliberate Boundaries [04:55 – 05:50 (55s)]

- • **Slide Content:** In-Scope: Modular PHP 8.2 & MySQL on Apache; 5 RBAC roles; OPD sequencing & Code128 barcode routing slips; testing bench with panic alerts; blood bank & specimen rejection QA; eCitizen Paybill 222222 billing. Out-of-Scope: Direct RS-232 machine serial cables; live national DHIS2 cloud push.

What to Say to the Panel (Speaker Script):
"To maintain engineering rigor, we established precise boundaries. In-scope deliverables include our 3-tier web stack, role-based access control across five hospital roles, barcode-enabled routing slips, doctor test ordering, panic alerts, blood bank safety, and cashless eCitizen settlement. We deliberately scoped out direct machine analyzer cabling and national DHIS2 cloud push for Phase 2, ensuring that our core system is fully self-reliant and immune to external hardware and internet downtime."

Defense Strategy & Examiner Tip:
Defend the scope as an intentional engineering decision for system independence and stability.

Slide 7: Justification & Significance: Multi-Stakeholder Impact [05:50 – 06:45 (55s)]

- • **Slide Content:** Administration: Stationery costs eliminated & revenue leakages sealed; Clinicians: Instant diagnostic history & clinical allergy banners; Lab Technologists: Clear worklists & automated reference ranges; Patients: Zero lost slips & faster care; Public Health: Real-time epidemiological data.

What to Say to the Panel (Speaker Script):
"The delivered system provides measurable value across every hospital stakeholder group. Hospital management eliminates stationery costs and stops revenue leakage. Clinicians gain instant access to patient diagnostic records and allergy warnings. Laboratory technologists benefit from automated reference ranges and 1-click reports. Patients experience zero lost slips and significantly reduced wait times. Finally, county

health authorities gain accurate epidemiological surveillance data."

Defense Strategy & Examiner Tip:
Remind the panel that clinical software must ultimately protect patient safety and diagnostic speed.

Slide 8: Resource Requirements & Budget: Economic Justification [06:45 – 07:40 (55s)]

- • **Slide Content:** Hardware: Core i5 workstation PC, thermal barcode printer, 650VA UPS; Software: PHP 8.2, MySQL 8.0, Apache/XAMPP (Zero recurring license fees); Capital Cost: KES 120,000; Training & Reserve: KES 50,000; Total Investment: KES 170,000 recovered via sealed leakage.

What to Say to the Panel (Speaker Script):
"From an economic standpoint, the system leverages open-source technologies—PHP 8.2, MySQL, and Apache—incurring zero annual software licensing costs compared to proprietary systems costing millions. Capital expenditures were limited to approximately KES 120,000 for server hardware and backup UPS units, plus KES 50,000 for staff training and user manuals. This modest investment is fully recovered within months through eliminated stationery and recovered billing leakages."

Defense Strategy & Examiner Tip:
Emphasize zero licensing fees—this is the most compelling argument for public hospital adoption.

Slide 9: Project Schedule & Agile Sprints [07:40 – 08:35 (55s)]

- • **Slide Content:** Agile Development Lifecycle across 4 iterative sprint cycles; Sprint 1: Analysis, inquiries & SRS; Sprint 2: Architecture, schema & RBAC; Sprint 3: CPOE, lab bench, blood bank & eCitizen; Sprint 4: 14 test cases, UAT evaluation (4.6/5.0) & deployment.

What to Say to the Panel (Speaker Script):
"Development was organized around the Agile methodology across four iterative two-week sprint cycles. Sprint 1 focused on requirements gathering and workflow observation. Sprint 2 established our 3-tier architecture, database schema, and RBAC authentication. Sprint 3 implemented core clinical modules—CPOE ordering, laboratory testing, blood bank, and eCitizen billing. Sprint 4 was dedicated to comprehensive functional testing, staff UAT evaluation, and deployment."

Defense Strategy & Examiner Tip:
Explain that Agile allowed mid-project enhancements like the blood bank crossmatch module based on staff

input.

Slide 10: Literature Review: Existing Systems Analysis [08:35 – 09:35 (60s)]

- • **Slide Content:** Manual MOH Paper: Cheap but fragile, error-prone, and search delays; OpenMRS & KenyaEMR: Robust for HIV/TB clinics, but heavy Java stack & lacks full lab bench accessioning; Commercial LIMS (LabWare): Feature-complete but millions in annual licenses; Cloud SaaS: Vulnerable to rural internet outages.

What to Say to the Panel (Speaker Script):

"Our literature review evaluated existing paradigms against public hospital realities. Manual paper is cheap but clinically dangerous. OpenMRS and KenyaEMR are successful national tools for HIV/TB care, but they are resource-heavy and lack full laboratory accessioning and bench workflows. Commercial solutions like LabWare cost millions of shillings in annual vendor licenses. Cloud-only SaaS solutions fail when rural internet goes down. Yala LIMS fills this exact niche: lightweight, zero-license, and 100% operational over the local intranet."

Defense Strategy & Examiner Tip:

Shows you understand the Kenyan healthcare IT landscape and where your system fits.

Slide 11: Literature Review: Gaps & Delivered Solution [09:35 – 10:35 (60s)]

- • **Slide Content:** Data Silos: Solved by shared relational database; Lack of Pre-Analytical QA: Solved by ISO 15189 specimen rejection logging; Internet Dependency: Solved by offline-capable LAN intranet on Apache/XAMPP; Cash Compliance: Solved by Government eCitizen Paybill 222222 integration.

What to Say to the Panel (Speaker Script):

"We identified four major operational gaps: inter-departmental communication silos, lack of pre-analytical specimen quality assurance, vulnerability to rural internet disruptions, and non-compliance with public cashless mandates. Yala LIMS directly addresses all four: it connects hospital departments over a local intranet, implements ISO 15189 specimen rejection controls, functions 100% without external internet, and reconciles Government eCitizen Paybill 222222 payments."

Defense Strategy & Examiner Tip:

Mention ISO 15189 quality standards and Kenya's cashless legal mandates.

Slide 12: Methodology & Development Tools Stack [10:35 – 11:35 (60s)]

- • **Slide Content:** Development Model: Agile SDLC; Backend: PHP 8.2 procedural architecture; Database: MySQL 8.0 with InnoDB foreign keys; Frontend: Bootstrap 5 & Chart.js; PDF Engine: FPDF library; IDE & VCS: Visual Studio Code & Git/GitHub.

What to Say to the Panel (Speaker Script):
"Our engineering stack was selected for reliability, speed, and low resource consumption. We utilized PHP 8.2 for backend business logic, MySQL 8.0 with InnoDB for relational persistence, Bootstrap 5 for a responsive medical UI, and FPDF for compiling digitally certified PDF reports. Code was managed using Git version control and developed within Visual Studio Code."

Defense Strategy & Examiner Tip:
Why procedural PHP instead of Laravel? It has lower overhead and hospital IT staff can easily maintain it.

Slide 13: System Analysis & Feasibility Study [11:35 – 12:35 (60s)]

- • **Slide Content:** Technical Feasibility: Stack operates smoothly on standard hospital core-i3/i5 PCs; Operational Feasibility: Screens strictly mirror physical MOH 204/240 register layouts; Economic Feasibility: High ROI via eliminated paper and captured billing; 4 User Profiles: Receptionist, Doctor, Lab Tech, Administrator.

What to Say to the Panel (Speaker Script):
"Our feasibility evaluation validated technical, operational, and economic viability. Technically, the system runs smoothly on standard workstation PCs over the local network. Operationally, staff adoption is rapid because our screens mirror familiar physical MOH registers. Economically, the cost savings and recovered billing justify deployment. We established four distinct user profiles enforcing strict Role-Based Access Control."

Defense Strategy & Examiner Tip:
Emphasize operational feasibility: mirroring physical MOH registers solved the staff adoption challenge.

Slide 14: System Requirements Modeling (Functional & Non-Functional) [12:35 – 13:35 (60s)]

- • **Slide Content:** Functional: Barcode patient registration, CPOE test ordering, automated panic alerts, certified PDF reports, eCitizen Paybill 222222 billing; Non-Functional: Bcrypt password hashing, MySQLi prepared statements, CSRF tokens, sub-1.5s LAN response time, automated daily SQL backups.

What to Say to the Panel (Speaker Script):
"Requirements modeling established functional and non-functional specifications. Functionally, the system

> *manages barcode patient registration, CPOE test ordering, automated panic alerting, certified PDF generation, and eCitizen billing. Non-functionally, we enforced enterprise-grade security via Bcrypt hashing, parameterized prepared statements, and CSRF token validation, maintaining sub-1.5 second page load times across the hospital LAN."*

> **Defense Strategy & Examiner Tip:**
> *Examiners will check whether your non-functional requirements were actually implemented in code.*

Slide 15: System Design & Architecture: The 3-Tier Blueprint [13:35 – 14:40 (65s)]

- • **Slide Content:** 3-Tier Client-Server Architecture: Presentation Tier (Bootstrap 5 & Chart.js), Application Tier (PHP 8.2 & FPDF), Data Persistence Tier (MySQL 8.0); Data Flow Modeling: Level 0 Context DFD and Level 1 DFD decomposing into 6 sub-processes.

> **What to Say to the Panel (Speaker Script):**
> *"Our architectural blueprint implements the classic 3-tier client-server architecture: a responsive Bootstrap 5 presentation layer, a modular PHP 8.2 application layer, and a normalized MySQL 8.0 data layer. Data flow modeling was decomposed through Level 0 and Level 1 DFDs, ensuring clean data separation between triage, clinical ordering, laboratory accessioning, and administrative auditing."*

> **Defense Strategy & Examiner Tip:**
> *Explain that the 3 tiers provide separation of concerns—changing the UI doesn't touch business logic or data.*

Slide 16: Database Schema & Security Controls [14:40 – 15:45 (65s)]

- • **Slide Content:** 12 normalized relational tables: patients, lab_requests, test_results, payments, blood_bank, sample_rejections, audit_log; Security: 100% prepared MySQLi queries, CSRF tokens, Bcrypt password hashing, immutable audit logging compliant with Data Protection Act 2019.

> **What to Say to the Panel (Speaker Script):**
> *"The relational database comprises 12 normalized tables with foreign key constraints. Core entities include patients, lab requests, test results, payments, blood bank inventory, and sample rejections. To guarantee data privacy and compliance with Kenya's Data Protection Act 2019, all database operations utilize parameterized prepared statements, while an immutable audit log captures user actions, IP addresses, and exact timestamps."*

Defense Strategy & Examiner Tip:
Mention Kenya's Data Protection Act 2019—it shows awareness of healthcare privacy law.

Slide 17: Core Modules Implemented: The 7 Operational Pillars [15:45 – 17:15 (90s)]

- • **Slide Content:** 1. Authentication & RBAC (index.php); 2. Outpatient Reception (add_patient.php); 3. Doctor CPOE Console (request_test.php); 4. Lab Workbench (enter_results.php); 5. eCitizen Cashless Billing (billing.php); 6. Blood Bank & Specimen QA; 7. MOH 706 Reports (reports.php).

What to Say to the Panel (Speaker Script):
"We successfully engineered seven operational modules: First, RBAC authentication. Second, outpatient reception with barcode slip printing. Third, doctor CPOE ordering with clinical allergy alerts. Fourth, the laboratory bench with panic toasts and certified PDF generation. Fifth, cashless billing reconciling Government eCitizen Paybill 222222 payments. Sixth, blood bank crossmatching and ISO 15189 sample rejection QA. Finally, 1-click MOH 706 monthly reporting and managerial KPI analytics."

Defense Strategy & Examiner Tip:
Spend 90 seconds here! This is the proof that you built a comprehensive, working medical system.

Slide 18: Testing & Operational Impact: Empirical Verification [17:15 – 18:30 (75s)]

- • **Slide Content:** 14 Functional Test Cases: 100% Pass Rate; User Acceptance Testing (UAT): 8 hospital staff scored 4.6 / 5.0 satisfaction; Missing Slips: Dropped from 12% to 0%; Transcription Errors: Eliminated; Monthly MOH 706 Compilation: Slashed from 2 full working days to 1 instant click; 100% staff endorsement.

What to Say to the Panel (Speaker Script):
"System validation combined functional verification and real-world evaluation. All 14 functional test cases achieved a 100% pass rate. User Acceptance Testing with eight active hospital staff yielded an overall satisfaction score of 4.6 out of 5.0. Empirically, lost result slips dropped from 12% to zero, transcription errors were eliminated, and compiling the monthly MOH report was reduced from two full working days to a single click. Most importantly, 100% of staff preferred the digital system over manual registers."

Defense Strategy & Examiner Tip:
The 4.6/5.0 UAT rating and the reduction from 2 days to 1 click are your strongest quantitative defenses.

Slide 19: Conclusions & Recommendations: The Future Roadmap [18:30 – 19:25 (55s)]

- • **Slide Content:** Success: 100% of objectives achieved; Deployment: System is operational on host PC at Yala Hospital; Future Phase 2: Direct RS-232/HL7 machine interfacing; Future Phase 3: Automated REST API push to National DHIS2 cloud; Scale-Up: Replicate across Siaya County Level 4 facilities.

What to Say to the Panel (Speaker Script):
"In conclusion, 100% of our project objectives were successfully achieved. The system is operational and deployed at Yala Sub-County Hospital. For future work, we recommend Phase 2 direct serial machine interfacing, Phase 3 automated cloud integration with Kenya's national DHIS2 platform, and scaling this open-source architecture across neighboring sub-county hospitals in Siaya County."

Defense Strategy & Examiner Tip:
Having a clear 3-phase roadmap shows maturity, vision, and realistic engineering planning.

Slide 20: References & Examination Invitation [19:25 – 20:00 (35s)]

- • **Slide Content:** Kenya National eHealth Policy (2016-2030); Kenya Data Protection Act 2019; MOH Laboratory SOPs; ISO 15189:2022 Medical Laboratories; Sommerville & Pressman Software Engineering textbooks.

What to Say to the Panel (Speaker Script):
"Our research is grounded in recognized academic literature, Ministry of Health laboratory operating procedures, Kenya's Data Protection Act 2019, and ISO 15189 standards for medical laboratory quality. Thank you for your time and attention. I now welcome questions, comments, and critique from the examining panel."

Defense Strategy & Examiner Tip:
Close with confidence and humility. Pause and prepare for the panel's first question.